



Sampling Methods in Research Methodology

A. Stage 1: Clearly Define Target Population

The first stage in the sampling process is to clearly define target population. Population is commonly related to the number of people living in a particular country.

B. Stage2: Select Sampling Frame

A sampling frame is a list of the actual cases from which sample will be drawn. The sampling frame must be representative of the population.

C. Stage 3: Choose Sampling Technique

Prior to examining the various types of sampling method, it is worth noting what is meant by sampling, along with reasons why researchers are likely to select a sample. Taking a subset from chosen sampling frame or entire population is called sampling. Sampling can be used to make inference about a population or to make generalization in relation to existing theory. In essence, this depends on choice of sampling technique.

In general, sampling techniques can be divided into two types:

- Probability or random sampling
- Non- probability or non- random sampling

Before choosing specific type of sampling technique, it is needed to decide broad sampling technique. Figure 2 shows the various types of sampling techniques.

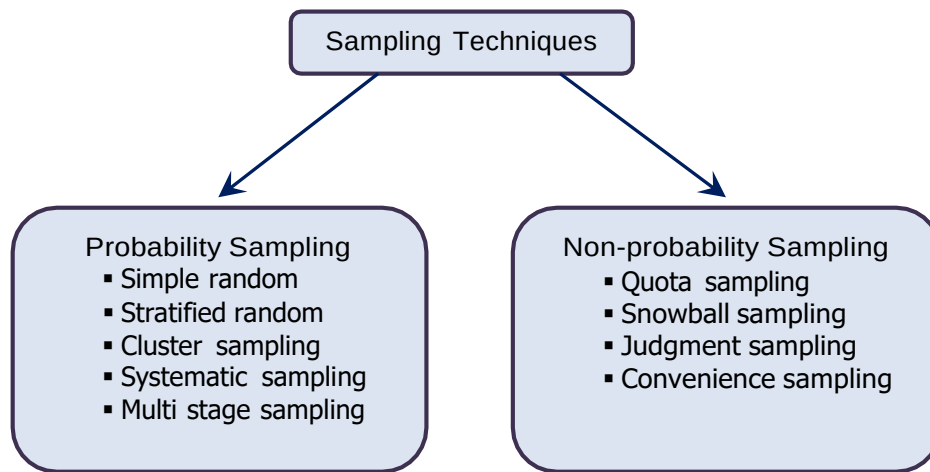


FIGURE 2: SAMPLING TECHNIQUES

1. Probability Sampling

Probability sampling means that every item in the population has an equal chance of being included in sample. One way to undertake random sampling would be if researcher was to construct a sampling frame first and then used a random number generation computer program to pick a sample from the sampling frame (Zikmund, 2002). Probability or random sampling has the greatest freedom from bias but may represent the most costly sample in terms of time and energy for a given level of sampling error (Brown, 1947).

1.1. Simple random sampling

The simple random sample means that every case of the population has an equal probability of inclusion in sample. Disadvantages associated with simple random sampling include (Ghauri and Gronhaug, 2005):

- ✓ A complete frame (a list of all units in the whole population) is needed;
- ✓ In some studies, such as surveys by personal interviews, the costs of obtaining the sample can be high if the units are geographically widely scattered;
- ✓ The standard errors of estimators can be high.

1.2. Systematic sampling

Systematic sampling is where every nth case after a random start is selected. Foreexample, if surveying a sample of consumers, every fifth consumer may be selected from your sample. The advantage of this sampling technique is its simplicity

1.3. Stratified random sampling

Stratified sampling is where the population is divided into strata (or subgroups) and a random sample is taken from each subgroup. A subgroup is a natural set of items. Subgroups might be based on company size, gender or occupation (to name but a few). Stratified sampling is often used where there is a great deal of variation within a population. Its purpose is to ensure that every stratum is adequately represented (Ackoff, 1953).

1.4. Cluster sampling

Cluster sampling is where the whole population is divided into clusters or groups. Subsequently, a random sample is taken from these clusters, all of which are used in the final sample (Wilson, 2010). Cluster sampling is advantageous for those researchers whose subjects are fragmented over large geographical areas as it saves time and money (Davis, 2005). The stages to cluster sampling can be summarized as follows:

- ✓ Choose cluster grouping for sampling frame, such as type of company or geographical region
- ✓ Number each of the clusters
- ✓ Select sample using random sampling

1.5. Multi-stage sampling

Multi-stage sampling (also known as multi-stage cluster sampling) is a more complex form of [cluster sampling](#) which contains two or more stages in sample selection. In simple terms, in multi-stage sampling large clusters of population are divided into smaller clusters in several stages in order to make primary data collection more manageable. It has to be acknowledged that multi-stage sampling is not as effective as true random sampling; however, it addresses certain disadvantages associated with true random sampling such as being overly expensive and time-consuming.

2. Non probability Sampling

Non probability sampling is often associated with case study research design and qualitative research. With regards to the latter, case studies tend to focus on small samples and are intended to examine a real life phenomenon, not to make statistical inferences in relation to the wider population (Yin, 2003). A sample of participants or cases does not need to be representative, or random, but a clear rationale is needed for the inclusion of some cases or individuals rather than others.

2.1. Quota sampling

Quota sampling is a non random sampling technique in which participants are chosen on

the basis of predetermined characteristics so that the total sample will have the same distribution of characteristics as the wider population (Davis, 2005).

2.2. Snowball sampling

Snowball sampling is a non random sampling method that uses a few cases to help encourage other cases to take part in the study, thereby increasing sample size. This approach is most applicable in small populations that are difficult to access due to their closed nature, e.g. secret societies and inaccessible professions (Breweton and Millward, 2001).

2.3. Convenience sampling

Convenience sampling is selecting participants because they are often readily and easily available. Typically, convenience sampling tends to be a favored sampling technique among students as it is inexpensive and an easy option compared to other sampling techniques (Ackoff, 1953). Convenience sampling often helps to overcome many of the limitations associated with research. For example, using friends or family as part of sample is easier than targeting unknown individuals.

2.4. Purposive or judgmental sampling

Purposive or judgmental sampling is a strategy in which particular settings persons or events are selected deliberately in order to provide important information that cannot be obtained from other choices (Maxwell, 1996). It is where the researcher includes cases or participants in the sample because they believe that they warrant inclusion.

Table 1 illustrates strengths and weaknesses associated with each respective sampling technique.

TABLE 1: STRENGTHS AND WEAKNESSES OF SAMPLING TECHNIQUES
SOURCE: (MALHOTRA AND BIRKS, 2006)

Technique	Strengths	Weaknesses
Convenience sampling	Least expensive, least time-consuming, most convenient	Selection bias, sample not representative, not recommended by descriptive or casual research
Judgment sampling	Low-cost, convenient, not time-consuming, ideal for exploratory research design	Does not allow generalization, subjective
Quota sampling	Sample can be controlled for certain characteristics	Selection bias, no assurance
Snowball sampling	Can estimate rare characteristics	Time-consuming
Simple random sampling	Easily understood, results projectable	Difficult to construct sampling frame, expensive, lower precision, no assurance of representativeness
Systematic sampling	Can increase representativeness, easier to implement than simple random sampling, sampling frame not always necessary	Can decrease representativeness
Stratified sampling	Includes all important sub-population, precision	Difficult to select relevant stratification variables, not feasible to stratify on many variables, expensive
Cluster sampling	Easy to implement, cost-effective	Imprecise, difficult to compute an interpret results

D. Stage 4: Determine Sample Size

In order to generalize from a random sample and avoid sampling errors or biases, a random sample needs to be of adequate size. What is adequate depends on several issues which often confuse people doing surveys for the first time. This is because what is important here is not the proportion of the research population that gets sampled, but the absolute size of the sample selected relative to the complexity of the population, the aims of the researcher and the kinds of statistical manipulation that will be used in data analysis. While the larger the sample the lesser the likelihood that findings will be biased does hold, diminishing returns can quickly set in when samples get over a specific size which need to be balanced against the researcher's resources (Gill et al., 2010). To put it bluntly, larger sample sizes reduce sampling error but at a decreasing rate. Several statistical formulas are available for determining sample size.

There are numerous approaches, incorporating a number of different formulas, for calculating the sample size for categorical data.

$$n = \frac{p(100-p)z^2}{E^2}$$

n is the required sample size

P is the percentage occurrence of a state or condition

E is the percentage maximum error required

Z is the value corresponding to level of confidence required